

Universidad Tecnológica El Retoño

IoT Application

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TIDSM5B

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Code

```
#include <SoftwareSerial.h>

SoftwareSerial SerialBT(0, 1);

const int sensorPin = A0;
int sensorValue = 0;
const int threshold = 100;

const int salidaPin = 2;
const int entradaPin = 3;
const int izq = 4;
const int der = 5;

const int led1 = 6;
const int led2 = 7;
const int led3 = 8;
const int led4 = 9;
const int led5 = 10;
const int led6 = 11;
const int led7 = 12;
const int led8 = 13;

bool automaticMode = true;

void setup() {
  Serial.begin(9600);
  SerialBT.begin(9600);

  pinMode(salidaPin, OUTPUT);
  pinMode(entradaPin, INPUT);
  pinMode(izq, OUTPUT);
  pinMode(der, OUTPUT);

  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
  pinMode(led3, OUTPUT);
  pinMode(led4, OUTPUT);
  pinMode(led5, OUTPUT);
  pinMode(led6, OUTPUT);
  pinMode(led7, OUTPUT);
  pinMode(led8, OUTPUT);
}
```

```
void loop() {
  digitalWrite(salidaPin, LOW);
  delayMicroseconds(2);
  digitalWrite(salidaPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(salidaPin, LOW);

  long duration = pulseIn(entradaPin, HIGH);
  int distance = duration * 0.034 / 2;

  int lectura = analogRead(A0);
  float voltage = lectura * (5.0 / 1023.0);
  float temperatura = voltage * 100.0;

  SerialBT.print("Distancia: ");
  SerialBT.print(distance);
  SerialBT.println(" cm");

  SerialBT.print("Temperatura: ");
  SerialBT.print(temperatura);
  SerialBT.println(" °C");

  sensorValue = analogRead(sensorPin);
  Serial.println(sensorValue);

  if (SerialBT.available()) {
    char inputBT = SerialBT.read();

    if (inputBT == '1') {
      automaticMode = true;
      analogWrite(izq, 0);
      analogWrite(der, 0);
      digitalWrite(led1, LOW);
      digitalWrite(led2, LOW);
      digitalWrite(led3, LOW);
      digitalWrite(led4, LOW);
      digitalWrite(led5, LOW);
      digitalWrite(led6, LOW);
      digitalWrite(led7, LOW);
      digitalWrite(led8, LOW);
    }
    else if (inputBT == '2') {
      automaticMode = false;
      digitalWrite(led1, HIGH);
    }
  }
}
```

```
digitalWrite(led2, HIGH);
digitalWrite(led3, HIGH);
digitalWrite(led4, HIGH);
digitalWrite(led5, HIGH);
digitalWrite(led6, HIGH);
digitalWrite(led7, HIGH);
digitalWrite(led8, HIGH);
}
else if (inputBT == '3') {
    automaticMode = false;
    analogWrite(izq, 0);
    analogWrite(der, 255);
}
else if (inputBT == '4') {
    automaticMode = false;
    analogWrite(izq, 255);
    analogWrite(der, 0);
}
else if (inputBT == '0') {
    automaticMode = false;
    analogWrite(izq, 0);
    analogWrite(der, 0);
    digitalWrite(led1, LOW);
    digitalWrite(led2, LOW);
    digitalWrite(led3, LOW);
    digitalWrite(led4, LOW);
    digitalWrite(led5, LOW);
    digitalWrite(led6, LOW);
    digitalWrite(led7, LOW);
    digitalWrite(led8, LOW);
}
}

if (automaticMode) {
    if (temperatura >= 40.0) {
        analogWrite(izq, 255);
        analogWrite(der, 0);
    }
    if (sensorValue <= threshold) {
        digitalWrite(led4, HIGH);
        digitalWrite(led5, HIGH);
    }
}
if (distance >= 6) {
    digitalWrite(led1, HIGH);
```

```

digitalWrite(led2, HIGH);
digitalWrite(led3, HIGH);
digitalWrite(led6, HIGH);
digitalWrite(led7, HIGH);
digitalWrite(led8, HIGH);
}
else {
  analogWrite(izq, 0);
  analogWrite(der, 0);
  digitalWrite(led1, LOW);
  digitalWrite(led2, LOW);
  digitalWrite(led3, LOW);
  digitalWrite(led4, LOW);
  digitalWrite(led5, LOW);
  digitalWrite(led6, LOW);
  digitalWrite(led7, LOW);
  digitalWrite(led8, LOW);
}
}
}
delay(500);
}

```

Blocks

